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Inventor(s)

Kruczek; Grzegorz et al.

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### HYDROGEN FUEL SYSTEM ARCHITECTURE WITH OXYGEN SEPARATION AND SUPERHEATING

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#### Abstract

A system comprising: a gas turbine having a combustion chamber and a turbine; a first heat exchanger configured to receive a first stream of compressed air and a stream of liquid hydrogen, wherein the first heat exchanger is configured to transfer heat between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen and a stream of gaseous hydrogen; a second heat exchanger configured to heat the stream of liquid oxygen to yield a stream of gaseous oxygen; and a third heat exchanger configured to receive a stream of flue gas from the turbine and at least one of the stream of gaseous hydrogen and the stream of gaseous oxygen, wherein the third heat exchanger is configured to transfer heat between the stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; wherein the system is configured to direct the stream of hydrogen and the stream of oxygen to the combustion chamber, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

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**Inventors:** Kruczek; Grzegorz (Wroclaw, PL), Ridouane; El Hassan (Cork, IE), Cao; Changmin (Cork, IE)

**Applicant:** Hamilton Sundstrand Corporation (Charlotte, NC)

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## **Background/Summary**

### **CROSS-REFERENCE TO RELATED APPLICATION AND PRIORITY CLAIM**

[0001] This application claims priority under 35 U.S.C. § 119 to European Patent Application No. 23461595.3 filed on May 23, 2023, which is hereby incorporated by reference in its entirety.

### **TECHNICAL FIELD**

[0002] The subject-matter disclosed herein relates to hydrogen fuel systems, and in particular to hybrid fuel cell gas turbine systems.

### **BACKGROUND**

[0003] There is an increasing pressure on the aerospace industry to switch the primary fuel for aircraft from hydrocarbons to more sustainable energy sources such as hydrogen. Hybrid fuel cell gas turbine systems, for example, comprise low-temperature fuel cells (e.g. proton exchange membrane fuel cells (PEMFCs)) that can increase fuel-to-electricity conversion efficiency and lower emissions of greenhouse gases. However, it is desirable to further increase efficiency and reduce emissions of these systems. The examples disclosed herein relate to such hydrogen fuel systems.

### **SUMMARY**

[0004] According to a first aspect, there is provided a system comprising: a gas turbine having a combustion chamber and a turbine; a first heat exchanger configured to receive a first stream of compressed air and a stream of liquid hydrogen, wherein the first heat exchanger is configured to transfer heat between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen and a stream of gaseous hydrogen; a second heat exchanger configured to heat the stream of liquid oxygen to yield a stream of gaseous oxygen; and a third heat exchanger configured to receive a stream of flue gas from the turbine and at least one of the stream of gaseous hydrogen and the stream of gaseous oxygen, wherein the third heat exchanger is configured to transfer heat between the stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; wherein the system is configured to direct the stream of hydrogen and the stream of oxygen to the combustion chamber, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

[0005] In the following, it should be understood that where the system is ‘configured to direct’ streams of fluids, the system comprises suitable conduits fluidly connected in such a manner as to facilitate the described path of the fluid stream.

[0006] The combustion chamber may be configured to combust the stream of superheated/gaseous hydrogen and the stream of superheated/gaseous oxygen. The temperature of the combustion of the streams of superheated/gaseous hydrogen and superheated/gaseous oxygen may be between 3000K to 4000K. The temperature of flue gases (produced by the combustion chamber) after leaving the turbine may be in the range of 500K to 2000K.

[0007] The term ‘superheated’ may be understood to mean that the fluid is at a temperature higher than its vaporisation point at absolute pressure. For example, a temperature of the stream of superheated hydrogen may be in the range 293K to 2000K, preferably between 400K to 500K (e.g. where the temperature of hydrogen should not exceed temperature of 500K in order to mitigate it escaping from the conduits/pipes of the system). A temperature of the stream of superheated

oxygen may be in the range 293K to 2000K, preferably between 600K to 800K.

[0008] The system may comprise a source of liquid hydrogen (e.g. a tank).

[0009] The second heat exchanger may be configured to receive a second stream of compressed air and configured to transfer heat from the second stream of compressed air to the stream of liquid oxygen to yield the stream of gaseous oxygen.

[0010] The system may comprise a compressor configured to generate the first stream of compressed air and the second stream of compressed air.

[0011] The third heat exchanger may be configured to receive one of the stream of gaseous hydrogen and the stream of gaseous oxygen; and the system may comprise a fourth heat exchanger configured to receive a stream of flue gas from the turbine and the other of the stream of gaseous hydrogen and the stream of gaseous oxygen, wherein the fourth heat exchanger is configured to transfer heat between the flue gas and the stream of gaseous hydrogen or the stream of gaseous oxygen to superheat the stream of hydrogen or the stream of oxygen; wherein the third heat exchanger and the fourth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; and wherein the system is configured to direct the stream of superheated oxygen and the stream of superheated hydrogen to the combustion chamber.

[0012] In other words, the system may comprise a single heat exchanger (a third heat exchanger) configured to transfer heat between a stream of flue gas and both a stream of gaseous hydrogen and a stream of gaseous oxygen in order to superheat the stream of hydrogen and the stream of oxygen; however, alternatively, the system may comprise one heat exchanger (a third heat exchanger) configured to transfer heat between a stream of flue gas and (a single) one of the streams of gaseous hydrogen and gaseous oxygen in order to superheat the stream of hydrogen or the stream of oxygen and another heat exchanger (a fourth heat exchanger) configured to transfer heat between a stream of flue gas and the other one of the streams of gaseous hydrogen and gaseous oxygen in order to superheat the stream of hydrogen or the stream of oxygen.

[0013] The system may comprise a fuel cell, and the system may be configured to: direct a first portion of the stream of gaseous hydrogen to the combustion chamber; and direct a second portion of the stream of gaseous hydrogen to the fuel cell. The second portion of the stream of gaseous hydrogen may be directed to the fuel cell via the second heat exchanger.

[0014] The system may be configured to direct the second stream of compressed air from the second heat exchanger to the fuel cell. Alternatively, the system may be configured to direct a stream of (optionally compressed) ambient air or bleed air to the fuel cell.

[0015] The system may be a hydrogen fuel system. The system may be a hybrid fuel cell gas turbine system comprising a fuel cell and a gas turbine. The system may be for providing propulsion to a vehicle, such as an aircraft.

[0016] The system may be configured to direct a stream of residual hydrogen from the fuel cell to the combustion chamber. However, the system may be configured to direct a stream of residual hydrogen from the fuel cell to the third heat exchanger (before it is passed to the combustion chamber) to superheat the stream of residual hydrogen. The system may be configured to then direct the stream of superheated residual hydrogen to the combustion chamber.

[0017] Alternatively, the system may comprise a fifth heat exchanger configured to receive a stream of flue gas from the turbine and the stream of residual hydrogen from the fuel cell, wherein the fifth heat exchanger is configured to transfer heat between the stream of flue gas and the stream of residual hydrogen to superheat the stream of residual hydrogen. The third heat exchanger, the fourth heat exchanger and the fifth heat exchanger may be arranged to receive the streams of flue gas from the turbine in series or in parallel. The system may be configured to direct the stream of superheated residual hydrogen to the combustion chamber.

[0018] In other words, the system may be configured to direct the stream of residual hydrogen from the fuel cell to the third heat exchanger to be superheated along with the first portion of the stream of gaseous hydrogen and/or the stream of gaseous oxygen; however, alternatively, the stream of

residual hydrogen may be directed to a dedicated additional (fifth) heat exchanger configured to superheat the stream of residual hydrogen.

[0019] The system may be configured to direct a stream of outlet air from the fuel cell to a turbine of the gas turbine.

[0020] The system may comprise a coolant circuit configured to absorb heat from (components of) the gas turbine and/or the fuel cell. For example, the coolant circuit may be configured to control the temperature of (e.g. cool) at least one of the fuel cell, an electric motor configured to provide power to the turbine, the turbine, the combustion chamber, and the compressor, as well as any other heat sources from other systems in the vehicle (e.g. aircraft). The coolant circuit may comprise an ambient heat exchanger configured to transfer heat from the coolant circuit to the ambient atmosphere. The coolant circuit may be a closed loop.

[0021] The coolant circuit may comprise the second heat exchanger. The second heat exchanger may be configured to transfer heat between the coolant circuit and at least one of the streams in the second heat exchanger (to thereby, for example, control the temperature of (e.g. heat) the streams in second heat exchanger).

[0022] The system may comprise a water recovery heat exchanger configured to receive a stream of flue gas from the turbine and a stream of ambient air, wherein the water recovery heat exchanger is configured to transfer heat from the stream of flue gas to the stream of ambient air to yield a stream of water. The water recovery heat exchanger may be arranged to receive the stream of flue gas from the turbine via other heat exchangers of the system. For example, the water recovery heat exchanger may be arranged to receive the stream of flue gas (directly) from at least one of the third heat exchanger, the fourth heat exchanger and the fifth heat exchanger. The system may be configured to direct the stream of water to a cooling system for the combustion chamber and/or the turbine. For example, a cooling system of the turbine may utilise the (liquid) water by injecting the water through inner channels of the turbine blades. In another example, a cooling system of the combustion chamber may utilise the water by injecting the water into the combustion chamber to lower the flame temperature.

[0023] The first heat exchanger may be configured to transfer heat between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen, a stream of gaseous hydrogen, and a stream of nitrogen.

[0024] The system may be configured to direct the stream of nitrogen to the second heat exchanger, wherein the second heat exchanger is configured to transfer heat between the stream of nitrogen and at least one of the streams in the second heat exchanger.

[0025] The system may be configured to direct the stream of nitrogen to a fire suppression system. For example, the stream of nitrogen may be used as a fire suppression agent (e.g. to put out fires) or a fire prevention agent (e.g. to cool down components at risk of catching fire).

[0026] The system may be configured to direct the stream of nitrogen to the source of liquid hydrogen (e.g. a tank) for temperature control (e.g. cooling) of the source of liquid hydrogen.

[0027] The system may be configured to direct the stream of gaseous nitrogen to vehicle electronics for temperature control (e.g. cooling) of the vehicle electronics. In other words, the vehicle comprising the hydrogen fuel system may utilise the stream of gaseous nitrogen to cool its electronic components and/or superconducting electronic components.

[0028] The system may be configured to direct the stream of gaseous nitrogen to turbine blades of (a turbine of) the gas turbine for temperature control (e.g. cooling) of the turbine blades.

[0029] In one example, the system may be configured to direct the stream of nitrogen to the source of liquid hydrogen for temperature control of the source of liquid hydrogen, from the source of liquid hydrogen to the second heat exchanger, wherein the second heat exchanger is configured to transfer heat between the stream of nitrogen and at least one of the streams in the second heat exchanger, and from the second heat exchanger to the fire suppression system.

[0030] According to a second aspect, there is provided a method comprising: directing a first

stream of compressed air and a stream of liquid hydrogen to a first heat exchanger; transferring heat, using the first heat exchanger, between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen and a stream of gaseous hydrogen; directing the stream of liquid oxygen to a second heat exchanger; transferring heat, using the second heat exchanger, to the stream of liquid oxygen to yield a stream of gaseous oxygen; directing a stream of flue gas from a turbine of a gas turbine, and at least one of the stream of gaseous hydrogen and the stream of gaseous oxygen, to a third heat exchanger; transferring heat, using the third heat exchanger, between the stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; and directing the stream of hydrogen and the stream of oxygen to a combustion chamber of the gas turbine, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

[0031] The method may comprise combusting, using the combustion chamber, the stream of gaseous hydrogen and the stream of gaseous oxygen.

[0032] The method may comprise directing the stream of liquid hydrogen to the first heat exchanger from a source of liquid hydrogen (e.g. a tank).

[0033] The method may comprise directing a second stream of compressed air to the second heat exchanger; and transferring heat, using the second heat exchanger, from the second stream of compressed air to the stream of liquid oxygen to yield the stream of gaseous oxygen.

[0034] The method may comprise generating, using a compressor, the first stream of compressed air and the second stream of compressed air. For example, the compressor may be used to generate a preliminary stream of compressed air that is split into the first stream and the second stream.

[0035] The method may comprise directing a stream of flue gas from the turbine of the gas turbine and one of the stream of gaseous hydrogen and the stream of gaseous oxygen to the third heat exchanger; directing a stream of flue gas from the turbine of the gas turbine and the other of the stream of gaseous hydrogen and the stream of gaseous oxygen to a fourth heat exchanger; wherein the third heat exchanger and the fourth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; transferring heat, using the third heat exchanger, between the stream of flue gas and the stream of gaseous hydrogen or the stream of gaseous oxygen to superheat the stream of hydrogen or the stream of oxygen; transferring heat, using the fourth heat exchanger, between the stream of flue gas and the stream of gaseous oxygen or the stream of gaseous hydrogen to superheat the stream of oxygen or the stream of hydrogen; and directing the stream of superheated oxygen and the stream of superheated hydrogen to a combustion chamber of the gas turbine.

[0036] The method may comprise directing a first portion of the stream of gaseous hydrogen from the first heat exchanger to the third heat exchanger or the combustion chamber; and directing a second portion of the stream of gaseous hydrogen from the first heat exchanger to a fuel cell, preferably via the second heat exchanger.

[0037] The method may comprise directing the second stream of compressed air from the second heat exchanger to the fuel cell. Alternatively, the system may be configured to direct a stream of (optionally compressed) ambient air or bleed air to the fuel cell.

[0038] The method may comprise directing a stream of residual hydrogen from the fuel cell to the combustion chamber.

[0039] Alternatively, the method may comprise directing a stream of residual hydrogen from the fuel cell to the third heat exchanger; transferring heat, using the third heat exchanger, between the stream of flue gas and the stream of residual hydrogen to superheat the stream of residual hydrogen; and directing the stream of superheated residual hydrogen to the combustion chamber.

[0040] Alternatively, the method may comprise directing a stream of flue gas from the turbine and the stream of residual hydrogen from the fuel cell to a fifth heat exchanger; wherein the third heat exchanger, the fourth heat exchanger and the fifth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; transferring heat, using the fifth heat

exchanger, between the stream of flue gas and the stream of residual hydrogen to superheat the stream of residual hydrogen; and directing the stream of superheated residual hydrogen to the combustion chamber.

[0041] In other words, the stream of residual hydrogen can be directed from the fuel cell directly to the combustion chamber; or it can be directed from the fuel cell to the third heat exchanger to be superheated along with the first portion of the stream of gaseous hydrogen and/or the stream of gaseous oxygen; or it can be directed from the fuel cell to an additional (fifth) heat exchanger configured to superheat the stream of residual hydrogen.

[0042] The method may comprise directing a stream of outlet air from the fuel cell to a turbine of the gas turbine.

[0043] The method may comprise providing a coolant circuit configured to absorb heat from the gas turbine and/or the fuel cell, wherein the coolant circuit comprises the second heat exchanger; and transferring heat, using the second exchanger, between the coolant circuit and at least one of the streams in the second heat exchanger.

[0044] The method may comprise directing a stream of flue gas from the turbine and a stream of ambient air to a water recovery heat exchanger, and transferring heat, using the water recovery heat exchanger, from the stream of flue gas to the stream of ambient air to yield a stream of water. The water recovery heat exchanger may be arranged to receive the stream of flue gas from the turbine via other heat exchangers of the system. For example, the water recovery heat exchanger may be arranged to receive the stream of flue gas (directly) from at least one of the third heat exchanger, the fourth heat exchanger and the fifth heat exchanger. The method may comprise directing the stream of water to a cooling system for the combustion chamber and/or the turbine. For example, a cooling system of the turbine may utilise the (liquid) water by injecting the water through inner channels of the turbine blades. In another example, a cooling system of the combustion chamber may utilise the water by injecting the water into the combustion chamber to lower the flame temperature.

[0045] The method may comprise transferring heat, using the first heat exchanger, between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen, a stream of gaseous hydrogen, and a stream of nitrogen.

[0046] The method may comprise directing the stream of nitrogen to the second heat exchanger; and transferring heat, using the second exchanger, between the stream of nitrogen and at least one of the streams in the second heat exchanger.

[0047] The method may comprise directing the stream of nitrogen to a fire suppression system. For example, the stream of nitrogen may be used as a fire suppression agent (e.g. to put out fires) or a fire prevention agent (e.g. to cool down components at risk of catching fire).

[0048] The method may comprise directing the stream of nitrogen to the source of liquid hydrogen (e.g. a tank) for temperature control (e.g. cooling) of the source of liquid hydrogen.

[0049] The method may comprise directing the stream of gaseous nitrogen to vehicle electronics for temperature control (e.g. cooling) of the vehicle electronics. In other words, the method may comprise directing the stream of gaseous nitrogen to (superconducting) electronic components of the vehicle (e.g. aircraft) comprising the hydrogen fuel system.

[0050] The method may comprise directing the stream of gaseous nitrogen to turbine blades of (a turbine of) the gas turbine for temperature control (e.g. cooling) of the turbine blades.

[0051] In one example, the method may comprise: directing the stream of nitrogen to the source of liquid hydrogen for temperature control of the source of liquid hydrogen; directing the stream of nitrogen from the source of liquid hydrogen to the second heat exchanger, wherein the second heat exchanger is configured to transfer heat between the stream of nitrogen and at least one of the streams in the second heat exchanger; and directing the stream of nitrogen from the second heat exchanger to the fire suppression system.

[0052] According to a third aspect, there is provided a system comprising: a gas turbine having a

combustion chamber and a turbine, the gas turbine being configured to receive a stream of gaseous oxygen and a stream of gaseous hydrogen, and to output a stream of flue gas; a water recovery heat exchanger configured to receive the stream of flue gas from the turbine, wherein the water recovery heat exchanger is configured to cool the stream of flue gas to yield a stream of water; wherein the system is configured to direct the stream of water to a cooling system for the combustion chamber and/or the turbine.

[0053] The system may comprise a superheating heat exchanger configured to receive the stream of flue gas from the turbine and at least one of the stream of gaseous hydrogen and the stream of gaseous oxygen, wherein the superheating heat exchanger is configured to transfer heat between the stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen.

[0054] The system may be configured to direct the stream of flue gas to the water recovery heat exchanger via the superheating heat exchanger. In other words, the water recovery heat exchanger may be arranged to receive the stream of flue gas (directly) from the superheating heat exchanger. In an example, the water recovery heat exchanger may be configured to receive the stream of flue gas and a stream of ambient air, and configured to transfer heat from the stream of flue gas to the stream of ambient air to yield the stream of water.

[0055] The system may be configured to direct the stream of hydrogen and the stream of oxygen to the combustion chamber, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

[0056] The system of the first or third aspect may comprise, combine with or interact with any or all elements of the system of the third or first aspect.

[0057] According to a fourth aspect there is provided a method comprising: combusting a stream of gaseous oxygen and a stream of gaseous hydrogen in a combustion chamber of a gas turbine; directing a stream of flue gas from a turbine of the gas turbine and a stream of ambient air to a water recovery heat exchanger, wherein the water recovery heat exchanger transfer heat from the stream of flue gas to the stream of ambient air to yield a stream of water; and directing the stream of water to a cooling system for the combustion chamber of the gas turbine and/or the turbine of the gas turbine.

[0058] The method may comprise: directing a stream of flue gas from the turbine and at least one of a stream of gaseous hydrogen and a stream of gaseous oxygen to a superheating heat exchanger; transferring heat, using the superheating heat exchanger, between the stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; directing the stream of flue gas from the superheating heat exchanger and a stream of ambient air to the water recovery heat exchanger; transferring heat, using the water recovery heat exchanger, from the stream of flue gas to the stream of ambient air to yield the stream of water; and directing the stream of hydrogen and the stream of oxygen to the combustion chamber, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

[0059] The method of the second or fourth aspect may comprise, combine with or interact with any or all elements of the method of the fourth or second aspect.

[0060] The foregoing features, elements and aspects may be combined in various combinations without exclusivity, unless expressly indicated otherwise. For example, any or all of the foregoing aspects of the disclosed methods may be applicable to and/or provided by the foregoing aspects of the disclosed systems, and vice versa.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

[0061] FIG. 1 shows a first example of a hydrogen fuel system architecture.

[0062] FIG. 2 shows a second example of a hydrogen fuel system architecture.

[0063] FIG. 3 shows a third example of a hydrogen fuel system architecture.

## DETAILED DESCRIPTION

[0064] There is an increasing pressure on the aerospace industry to switch the primary fuel used for aircraft from hydrocarbons to more sustainable energy sources such as hydrogen. This, however, brings many challenges. The systems described herein aim to take advantage of the new technologies of cryogenic compressed hydrogen and liquid hydrogen storage. In particular, the new systems described herein utilise the chill that is associated with cryogenic fluid.

[0065] The new systems described herein envision a way to use liquid H<sub>2</sub> onboard an aircraft as an additional heat sink to separate oxygen and nitrogen from part of the air. More specifically the systems use a new architecture of hybrid gas turbine and fuel cell operation with an oxygen and nitrogen separation process.

[0066] In order to reach a manageable H<sub>2</sub> volume, it must either be highly compressed and cooled (between 300-850 bar and <78K) or cryogenically cooled to liquid state (~-253° C.=~20K). The use of liquid hydrogen (LH) is more promising in terms of weight and volume when compared to compressed hydrogen (CH) and represents the only way to enable hydrogen-based short and medium range and bigger type of aircrafts.

[0067] The low storage temperature of liquid hydrogen fuel allows it to be used to cool air to condense oxygen and effectively separate it from nitrogen. Advantageously, the separated oxygen can be combusted with hydrogen in oxy-fuel combustion at very high temperatures which increases the efficiency of gas turbine. The absence of nitrogen in the combustion chamber avoids the generation of harmful emissions. Furthermore, the separated nitrogen can be used as a cooling medium (e.g. for cooling one or more of a tank of liquid hydrogen fuel, electronics, superconducting electronics, turbine blades, etc), as a fire suppression agent, and/or as a neutral gas for fire protection.

[0068] Thus, the described configurations allow for the use of high efficiency oxy-fuel combustion and additionally the production of nitrogen which can be used as a fire suppressant or fire protection gas. Part of the hydrogen that was used to condense oxygen is combusted in a combustion chamber along with oxygen and the rest, along with a second portion of air, may optionally also be used in fuel cell for a hybrid fuel cell gas turbine configuration.

[0069] The separation process of the new systems described herein are possible due to the fact that the oxygen boiling point is at 90.2K which is significantly higher than temperature of liquid hydrogen (~20K). Moreover, the nitrogen boiling point is at 77.4K which gives a range of temperatures where liquid oxygen and gaseous nitrogen exist (77.4K to 90.2K).

[0070] For improved functioning of the hydrogen-engine, or improved functioning of a fuel cell, the hydrogen fuel needs to reach higher temperatures (>0° C.=273K). The described configurations provide a means to increase the hydrogen temperature, whilst simultaneously producing high purity oxygen (e.g. comprising at least 95 mol. % oxygen) for oxy-fuel combustion.

[0071] An architecture for a hydrogen fuel system **100** is shown in FIG. 1. The hydrogen fuel system **100** shown in FIG. 1 is a hybrid fuel cell gas turbine system comprising a fuel cell **102** and a gas turbine engine **110**. In the example shown in FIG. 1, the hydrogen fuel system **100** is arranged in an aircraft (not shown) to provide electricity and propulsion to the aircraft.

[0072] As shown in FIG. 1, the hydrogen fuel system **100** comprises a tank **104** as a source of liquid hydrogen LH. As also shown in FIG. 1, the hydrogen fuel system **100** comprises a compressor **112** configured to compress air A, such as ambient air or bleed air, to provide a stream of compressed air CA.

[0073] The hydrogen fuel system **100** comprises at least a first (oxygen condensing) heat exchanger



HX1, a second (oxygen heating/evaporating) heat exchanger HX2, and a third (superheating) heat exchanger HX3.

[0074] The first heat exchanger HX1 is configured to receive a first stream of compressed air CA1 (a first portion of the stream of compressed air CA from the compressor **112**) and a stream of liquid hydrogen LH from the tank **104**. The first heat exchanger HX1 is configured to transfer heat between the first stream of compressed air CA1 and the stream of liquid hydrogen LH to yield a stream of liquid oxygen LO and a stream of gaseous hydrogen GH.

[0075] Valves (not shown) regulate the rate of flow of the first stream of compressed air CA1 and rate of flow of the stream of liquid hydrogen LH to the first heat exchanger HX1. The rate of flow of the first stream of compressed air CA1 is selected based on an oxygen demand of the gas turbine engine **110**. The rate of flow of the stream of liquid hydrogen LH is selected based on a combined hydrogen demand of the fuel cell **102** and the gas turbine engine **110**.

[0076] The second heat exchanger HX2 is configured to receive at least the stream of liquid oxygen LO and a second stream of compressed air CA2 (a second portion of the stream of compressed air CA from the compressor **112**). The second heat exchanger HX2 is configured to transfer heat between the second stream of compressed air CA2 and the stream of liquid oxygen LO to yield a stream of gaseous oxygen GO.

[0077] The gas turbine engine **110** comprises a combustion chamber **114** and a turbine **116**. The hydrogen fuel system **100** is configured to provide the combustion chamber **114** with a stream of oxygen and a stream of hydrogen (discussed further below). The hydrogen fuel system **100** is further configured to direct a stream of (hot) flue gas FG, generated from the combustion of the oxygen and hydrogen within the combustion chamber **114**, from the combustion chamber **114** to the turbine **116**. As the combustion chamber **114** utilises oxy-fuel combustion, the hot flue gas FG is substantially composed of superheated steam.

[0078] In the example shown in FIG. **1**, the third heat exchanger HX3 is (a hydrogen superheating heat exchanger) configured to receive the stream of flue gas FG from the turbine **116** and a first stream of gaseous hydrogen GH1 (a first portion of the stream of gaseous hydrogen GH from the first heat exchanger HX1). The third heat exchanger HX3 is configured to transfer heat between the stream of flue gas FG and the first stream of gaseous hydrogen GH1 to superheat the stream of hydrogen (e.g. to yield a stream of superheated hydrogen SH). The system **100** is configured to direct the stream of superheated hydrogen SH and the stream of gaseous oxygen GO to the combustion chamber **114**.

[0079] In the alternative example shown in FIG. **2**, the third heat exchanger HX3 is instead (an oxygen superheating heat exchanger) configured to receive the stream of flue gas FG from the turbine **116** and the stream of gaseous oxygen GO. The third heat exchanger HX3 is configured to transfer heat between the stream of flue gas FG and the stream of gaseous oxygen GO to superheat the stream of oxygen (e.g. to yield a stream of superheated oxygen SO). The system **100** is configured to direct the stream of superheated oxygen GO and the first stream of gaseous hydrogen GH1 to the combustion chamber **114**.

[0080] It should be understood that the hydrogen fuel system **100** of FIG. **1** may comprise an additional (fourth) heat exchanger similar to the third heat exchanger HX3 of the system **100** of FIG. **2** (e.g. an oxygen superheating heat exchanger, configured to receive a stream of flue gas FG and the stream of gaseous oxygen GO, and to transfer heat between the two). Said fourth heat exchanger may be configured in parallel with the third heat exchanger HX3, so as to receive a portion of the stream of flue gas FG from the turbine **116**, or in series with the third heat exchanger HX3 to receive the stream of flue gas FG upstream of or downstream from the third heat exchanger HX3.

[0081] In the further alternative example shown in FIG. **3**, the third heat exchanger HX3 is (a hydrogen-oxygen superheating heat exchanger) configured to receive at least a stream of flue gas FG from the turbine **116**, a first stream of gaseous hydrogen GH1 (a first portion of the stream of

gaseous hydrogen GH from the first heat exchanger HX1), and the stream of gaseous oxygen GO. In this example, the third heat exchanger HX3 is configured to transfer heat between the stream of flue gas FG and the first stream of gaseous hydrogen GH1 and the stream of gaseous oxygen GO to superheat the stream of hydrogen and the stream of oxygen (e.g. to yield a stream of superheated hydrogen SH and a stream of superheated oxygen SO). The system **100** is configured to direct the stream of superheated hydrogen SH and the stream of superheated oxygen SO to the combustion chamber **114**.

[0082] It will be appreciated that in the example shown in FIG. **3** the first stream of gaseous hydrogen GH1 and the stream of gaseous oxygen GO may optionally be mixed prior to entering the third heat exchanger HX3.

[0083] In the examples shown in FIGS. **1-3**, the hybrid system **100** is configured to direct a second stream of gaseous hydrogen GH2 (a second portion of the stream of gaseous hydrogen GH from the first heat exchanger HX1) to the fuel cell **102**.

[0084] The fuel cell **102** is configured to receive the second stream of gaseous hydrogen GH2 from the first heat exchanger HX1 and the second stream of compressed air CA2 from the second heat exchanger HX2. However, compressed air may be supplied to the fuel cell **102** from other sources.

[0085] A conversion ratio of the fuel cell **102** may be in the range of 0.8-0.98, such that a stream of residual hydrogen RH is output from the fuel cell **102**. As shown in the example in FIG. **1**, the hydrogen fuel system **100** may be configured to direct the stream of residual hydrogen RH (e.g. hydrogen that is not utilised in the fuel cell reaction) to the combustion chamber **114**. Alternatively (as shown in FIG. **3**), the hydrogen fuel system **100** may be arranged to direct the stream of residual hydrogen RH to the third heat exchanger HX3 to superheat the stream of residual hydrogen (e.g. to yield a stream of superheated residual hydrogen SRH), and then to direct the stream of superheated residual hydrogen SRH to the combustion chamber **114** to be used in combustion.

[0086] In another alternative example not shown in the Figures, the hydrogen fuel system **100** comprises a fifth heat exchanger, which is (a residual hydrogen superheating heat exchanger) configured to receive a stream of flue gas FG from the turbine and the stream of residual hydrogen RH from the fuel cell **102**, and configured to transfer heat between the stream of flue gas FG and the stream of residual hydrogen RH to superheat the residual hydrogen (e.g. to yield the stream of superheated residual hydrogen SRH). Said fifth heat exchanger may be arranged in parallel with the third heat exchanger HX3 and fourth heat exchanger, so as to receive a portion of the stream of flue gas FG from the turbine **116**, or in series with the third heat exchanger HX3 and fourth heat exchanger to receive the stream of flue gas FG upstream of or downstream from the third heat exchanger HX3 or fourth heat exchanger.

[0087] The hydrogen fuel system **100** may be configured to direct a stream of outlet air OA (e.g., air that is not utilised in the fuel cell reaction) from the fuel cell **102** to the turbine **116** of the gas turbine engine **110**.

[0088] As shown in FIGS. **1-3**, the hydrogen fuel system **100** may comprise a water recovery heat exchanger WHX configured to receive a stream of flue gas FG from the turbine **116** and a stream of ambient air AA, wherein the water recovery heat exchanger WHX is configured to transfer heat from the stream of flue gas FG to the stream of ambient air AA to yield a stream of water LW. The water recovery heat exchanger WHX may be arranged to receive the stream of flue gas FG from the turbine **116** via other heat exchangers of the system **100**. For example, as shown in FIG. **1**, the water recovery heat exchanger WHX is arranged to receive the stream of flue gas (directly) from the third (hydrogen superheating) heat exchanger HX3. As shown in FIG. **2**, the water recovery heat exchanger WHX is arranged to receive the stream of flue gas (directly) from the third (oxygen superheating) heat exchanger HX3. As shown in FIG. **3**, the water recovery heat exchanger WHX is arranged to receive the stream of flue gas (directly) from the third (hydrogen-oxygen superheating) heat exchanger HX3.

[0089] In hydrogen fuel systems having fourth heat exchangers and fifth heat exchangers as described above, the water recovery heat exchanger WHX may be arranged to receive the stream of flue gas from at least one of the third heat exchanger, the fourth heat exchanger and the fifth heat exchanger. The third heat exchanger, fourth heat exchanger and fifth heat exchanger may be arranged in the flue gas stream in series or in parallel.

[0090] The system may be configured to direct the stream of water to a cooling system for the combustion chamber and/or the turbine. For example, a cooling system of the turbine may utilise the (liquid) water by injecting the water through inner channels of the turbine blades. In another example, a cooling system of the combustion chamber may utilise the water by injecting the water into the combustion chamber to lower the flame temperature.

[0091] As shown in FIGS. 1-3, the hybrid system **100** comprises a coolant circuit CC configured to absorb heat from components of the gas turbine engine **110** and/or the fuel cell **102** into a working fluid within the coolant circuit CC. For example, as shown in FIGS. 1-3, the coolant circuit CC is configured to convey a flow of working fluid to control the temperature of (e.g. cool) the fuel cell **102** and an electric motor **118** coupled to the gas turbine engine **110**.

[0092] The coolant circuit CC comprises an ambient heat exchanger HXA. The ambient heat exchanger HXA is configured to transfer heat between the working fluid of the coolant circuit CC and the ambient atmosphere (e.g. to reject heat from the coolant circuit CC to the ambient atmosphere to cool the working fluid).

[0093] In the examples shown in FIGS. 1-3, the coolant circuit CC comprises the second heat exchanger HX2. Accordingly, the second heat exchanger HX2 is configured to transfer heat between the coolant circuit CC and at least one of the fluid streams in the second heat exchanger HX2. For example, the second heat exchanger HX2 may be configured to transfer heat between the coolant circuit CC and one or more of the second (portion of the) stream of gaseous hydrogen GH2, the stream of liquid oxygen LO, the second stream of compressed air CA2, and the stream of gaseous nitrogen GN.

[0094] In addition to generating a stream of liquid oxygen LO from the first stream of compressed air CA1, the first heat exchanger HX1 may be configured to generate a stream of gaseous nitrogen GN from the first stream of compressed air CA1. In other words, the first heat exchanger HX1 may be configured to transfer heat between the first stream of compressed air CA1 and the stream of liquid hydrogen LH to yield a stream of liquid oxygen LO, a stream of gaseous hydrogen GH, and a stream of gaseous nitrogen GN.

[0095] As shown in FIGS. 1-3, the system **100** is configured to direct the stream of gaseous nitrogen GN to the second heat exchanger HX2, and the second heat exchanger HX2 is configured to transfer heat between the stream of gaseous nitrogen GN and at least one of the other streams (e.g. the stream of liquid oxygen LO and/or the second stream of compressed air CA2) in the second heat exchanger HX2.

[0096] Alternatively, or additionally, the stream of gaseous nitrogen GN may be directed to a fire suppression system **108** (e.g. for use as a fire suppression or prevention agent). For this use, the stream of gaseous nitrogen GN should contain less than 15 mol. % oxygen, which is the maximum oxygen content that can still extinguish a fire. In various embodiments, the stream of gaseous nitrogen GN may contain less than 5 mol. % oxygen, or less than 1 mol. % oxygen.

[0097] Alternatively, or additionally, the stream of gaseous nitrogen GN may be directed to the tank **104** of liquid hydrogen to provide/aid in temperature control (e.g. cooling) of the tank **104**. Similarly, the stream of gaseous nitrogen GN may be directed to (superconducting) electronic components of the aircraft, and/or directed to the turbine blades of the turbine, to provide/aid in temperature control (e.g. cooling) of these components.

[0098] For example, the system **100** could be configured to direct the stream of gaseous nitrogen GN from the first heat exchanger HX1 to the tank **104** for temperature control of the tank **104**, and then from the tank **104** to the second heat exchanger HX2, wherein the second heat exchanger HX2

is configured to transfer heat between the stream of gaseous nitrogen GN and at least one of the streams (e.g. the stream of liquid oxygen LO and/or the second stream of compressed air CA2) in the second heat exchanger HX2, and then from the second heat exchanger HX2 to the fire suppression system **108**.

[0099] A method of operating the hydrogen fuel system **100** will now be described with reference to the Figures.

[0100] Firstly, the method comprises directing a first stream of compressed air CA1 to the first heat exchanger HX1. The compressor **112** may be used to generate a stream of compressed air CA by compressing a stream of air A (for example) to approximately 2.5 bar absolute pressure. Depending on the ambient conditions, the temperature of the compressed air after leaving the compressor **112** may range from 30-220° C. Next, most (approximately 80%) of the stream of compressed air CA is diverted to an environmental control system **120**. The remaining stream of compressed air is split into a first stream of compressed air CA1 (approximately 40% of the stream of compressed air CA) and a second stream of compressed air CA2 (approximately 60% of the stream of compressed air CA).

[0101] In addition to directing the first stream of compressed air CA1 to the first heat exchanger HX1, a stream of liquid hydrogen LH is directed to the first heat exchanger HX1. The stream of liquid hydrogen LH may be provided from a source of liquid hydrogen such as a tank **104**. The hydrogen stored within has been cryogenically cooled to liquid state (~-253° C.=~20K).

[0102] To raise the temperature of the stream of liquid hydrogen LH and evaporate it to yield a stream of gaseous hydrogen GH, the first heat exchanger HX1 transfers heat between the first stream of compressed air CA1 and the stream of liquid hydrogen LH. The first stream of compressed air CA1 is thereby cooled to condense the oxygen in the compressed air and a stream of liquid oxygen LO is separated out, leaving behind a stream of gaseous nitrogen GN. The outlet temperature of the stream of liquid oxygen LO and the stream of gaseous nitrogen GN is in the range of 77.4-90.2K.

[0103] After leaving first heat exchanger HX1, the stream of gaseous hydrogen GH is split into a first portion and a second portion: a first (portion of the) stream of gaseous hydrogen GH1 is directed to the combustion chamber **114** via the third heat exchanger HX3, and a second (portion of the) stream of gaseous hydrogen GH2 is directed to the fuel cell **102** via the second heat exchanger HX2.

[0104] After leaving the first heat exchanger HX1, the stream of liquid oxygen LO and the stream of gaseous nitrogen GN are directed to a second heat exchanger HX2. The second stream of compressed air CA2 is also directed to the second heat exchanger HX2. As shown in FIGS. **1-3**, the second heat exchanger HX2 also forms part of a coolant circuit CC.

[0105] The coolant circuit CC is arranged in a heat exchange relationship with components of the gas turbine engine and components of the fuel cell, such that the working fluid circulating in the coolant circuit CC absorbs heat from these components. Conventionally, this heat in the coolant circuit CC is rejected/transferred, in the ambient heat exchanger HXA, to a stream of ambient air AA. However, in the described systems, the heat in the coolant circuit CC can be utilised in the second heat exchanger HX2. Thus, in the second heat exchanger HX2, heat is transferred between the coolant circuit CC and the streams of liquid oxygen LO, gaseous hydrogen GH2, compressed air CA2, and gaseous nitrogen GN in order to yield a stream of gaseous oxygen GO and to control and/or stabilise the temperature of each of the streams of gaseous hydrogen GH2, compressed air CA2, and gaseous nitrogen GN depending on its particular requirements.

[0106] For example, the streams of liquid oxygen LO and gaseous nitrogen GN may be used to cool the temperature of the compressed air CA2, because, to be used in the fuel cell **102**, the air optimally should be in the operating range of 0-95° C. Then, the stream of gaseous oxygen GO in the second heat exchanger HX2 may be heated by the coolant circuit CC so that it is at an optimal temperature for combustion.

[0107] With reference to FIG. 1, the stream of hot flue gas FG (from the turbine **116**) and the first stream of gaseous hydrogen GH1 (from the first heat exchanger HX1) are directed to the third heat exchanger HX3. Thus, in the third heat exchanger HX3, heat is transferred between the stream of flue gas FG and the stream of gaseous hydrogen GH1 in order to superheat the stream of hydrogen. Then, the stream of superheated hydrogen SH (from the third heat exchanger HX3) and the stream of gaseous oxygen GO (from the second heat exchanger HX2) are directed to the combustion chamber **114** to be combusted.

[0108] With reference to FIG. 2, the stream of hot flue gas FG (from the turbine **116**) and the stream of gaseous oxygen GO (from the second heat exchanger HX2) are directed to the third heat exchanger HX3. Thus, in the third heat exchanger HX3, heat is transferred between the stream of flue gas FG and the stream of gaseous oxygen GO in order to superheat the stream of oxygen. Then, the stream of gaseous hydrogen GH1 (from the first heat exchanger HX1) and the stream of superheated oxygen SO (from the third heat exchanger HX3) are directed to the combustion chamber **114** to be combusted.

[0109] With reference to FIG. 3, the stream of hot flue gas FG (from the turbine **116**), the stream of gaseous oxygen GO (from the second heat exchanger HX2), and the first stream of gaseous hydrogen GH1 (from the first heat exchanger HX1) are directed to the third heat exchanger HX3. Thus, in the third heat exchanger HX3, heat is transferred between the stream of flue gas FG, and the stream of gaseous oxygen GO and the stream of gaseous hydrogen GH1, in order to superheat the stream of hydrogen and the stream of oxygen. Then, the superheated streams SH, SO are directed to the combustion chamber **114** to be combusted.

[0110] In examples comprising a third heat exchanger HX3 and a fourth heat exchanger (not shown), a stream of flue gas FG and the first stream of gaseous hydrogen GH1 are directed to the third heat exchanger, and a stream of flue gas FG and the stream of gaseous oxygen GO are directed to the fourth heat exchanger. The stream of flue gas FG may be directed to the fourth heat exchanger either from the turbine **116** or from the third heat exchanger HX3. Then, the stream of superheated hydrogen SH (from the third heat exchanger HX3) and the stream of superheated oxygen SO (from the fourth heat exchanger) are directed to the combustion chamber **114** to be combusted.

[0111] Thus, the gaseous oxygen GO and the gaseous hydrogen GH may be combusted in the combustion chamber **114** to yield a stream of (hot) flue gas FG. The stream of flue gas FG is directed to the turbine **116** to be utilised by the turbine **116** to generate mechanical and/or electrical power.

[0112] After leaving the second heat exchanger HX2, the stream of compressed air CA2 is directed to the fuel cell **102**. The fuel cell **102** utilises the stream of compressed air CA2 and the second stream of gaseous hydrogen GH2 to generate electricity. The fuel cell conversion ratio may be in the range of 0.8-0.98, such that there is residual hydrogen from the fuel cell **102** that is not used in the fuel cell conversion process. This stream of residual hydrogen RH is directed to the combustion chamber **114** to be used in combustion.

[0113] Alternatively (as shown in FIG. 3), the stream of residual hydrogen RH is first directed to the third heat exchanger HX3 to superheat the stream of residual hydrogen RH, and then the stream of superheated residual hydrogen SRH is directed to the combustion chamber **114** to be used in combustion. In another alternative example (not shown), the stream of residual hydrogen RH is first directed to a fifth heat exchanger to superheat the stream of residual hydrogen RH, and then the stream of superheated residual hydrogen SRH is directed to the combustion chamber **114** to be used in combustion.

[0114] Similarly, air that is not used as part of the fuel cell conversion process is directed as a stream of outlet air OA to the turbine **116**.

[0115] As shown in FIGS. 1-3, a stream of flue gas FG from the turbine **116** that has (at least) passed through the third heat exchanger HX3 may be directed to the water recovery heat exchanger

WHX. Using the water recovery heat exchanger WHX, heat is transferred (e.g. rejected) from the stream of flue gas to the stream of ambient air to yield a stream of water out of the flue gas (which is, as mentioned above, predominantly steam). In hydrogen fuel systems **100** having fourth heat exchangers and fifth heat exchangers as described above, the water recovery heat exchanger WHX may be arranged to receive the stream of flue gas from at least one of the third heat exchanger, the fourth heat exchanger and the fifth heat exchanger. The third heat exchanger, fourth heat exchanger and fifth heat exchanger may be arranged in the flue gas stream FG in series or in parallel.

[0116] The stream of water LW may be directed to a cooling system (not shown) for the combustion chamber **114** and/or the turbine **116**. For example, a cooling system of the turbine **116** may utilise the (liquid) water by injecting the water through inner channels of the turbine blades. In another example, a cooling system of the combustion chamber **114** may utilise the water by injecting the water into the combustion chamber **114** to lower the flame temperature.

[0117] After leaving the second heat exchanger HX2, the stream of gaseous nitrogen GN is directed to a fire suppression system **108** to be stored for use as a cooling medium (e.g. for cooling one or more of a tank of liquid hydrogen fuel, electronics, superconducting electronics, turbine blades, etc), as a fire suppressant, and/or as a fire prevention agent. Thus, an aircraft having the described hydrogen fuel system may not require a separate source of nitrogen for fire suppression, lowering the overall weight of the aircraft.

[0118] In any of the examples shown in FIGS. **1** to **3**, different types of heat exchangers HX may be used in this configuration based on the hydrogen and oxygen phase at its entrance (liquid, gas or a two-phase flow). Under different conditions a system controller may be configured to modify the system operating points, for example by controlling the liquid hydrogen, air or working fluid flow entering the heat exchangers HX through a bypass valve (not shown in the Figures). In some examples, the amount of working fluid or ambient air flowing into the fourth heat exchanger HXA may be controlled through a (bypass) valve or radiator baffle (not shown in the Figures) to ensure a proper working fluid temperature under different ambient conditions. Depending on the configuration, the heat exchangers HX may include components for moisture removal (such as a condenser, water separator and re-heater) to protect from possible freezing issues.

[0119] In one example, the first heat exchanger HX1 could instead be configured to produce a stream of liquid nitrogen instead of a stream of gaseous nitrogen, where the stream of liquid nitrogen is then heated in the second heat exchanger HX2 to produce the stream of gaseous nitrogen GN.

[0120] The examples described herein and discussed with reference to FIGS. **1** to **3** provide a more sustainable architecture and use of oxy-fuel combustion. As a result of combusting pure oxygen and pure hydrogen, no harmful compounds such as nitrogen oxides are produced in the combustion chamber **114** and thus the described examples provide for a higher gas turbine efficiency.

Moreover, superheating the pure oxygen and pure hydrogen before combustion (utilising otherwise waste heat energy) further improves gas turbine efficiency.

[0121] Additionally, some examples are configured to produce a source of nitrogen which in turn can be used as a fire suppression agent or fire protection agent. There is therefore no need for compressed nitrogen tanks for fire suppression, which in turn results in a lower aircraft weight. Due to the cryogenic temperature of the air, air pollutants are in the solid state, meaning that the pollutant (CO.sub.2, hydrocarbons, aromatics, etc.) separation process is significantly easier.

[0122] As can be seen in the Figures, since only one compressor **112** is needed, this results in a lower aircraft weight. Since only one turbine **116** is needed, this also results in a lower aircraft weight. Thus, the examples described herein also provide a simpler gas turbine design, that is cheaper, more reliable, has a higher energy efficiency and lower maintenance costs.

[0123] The systems described above may also comprise additional auxiliary equipment such as pumps, solid separator, dehumidifiers, valves, splitters etc, which are not shown in the Figures.

[0124] Although this disclosure has been described in terms of preferred examples, it should be

understood that these examples are illustrative only and that the claims are not limited to those examples. Those skilled in the art will be able to make modifications and alternatives in view of the disclosure which are contemplated as falling within the scope of the appended claims.

## Claims

1. A system comprising: a gas turbine having a combustion chamber and a turbine; a first heat exchanger configured to receive a first stream of compressed air and a stream of liquid hydrogen, wherein the first heat exchanger is configured to transfer heat between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen and a stream of gaseous hydrogen; a second heat exchanger configured to heat the stream of liquid oxygen to yield a stream of gaseous oxygen; and a third heat exchanger configured to receive a first stream of flue gas from the turbine and at least one of the stream of gaseous hydrogen or the stream of gaseous oxygen, wherein the third heat exchanger is configured to transfer heat between the first stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; wherein the system is configured to direct the stream of hydrogen and the stream of oxygen to the combustion chamber, wherein at least one of the stream of hydrogen or the stream of oxygen is superheated.
2. The system of claim 1, wherein the third heat exchanger is configured to receive one of the stream of gaseous hydrogen and the stream of gaseous oxygen; and wherein the system further comprises a fourth heat exchanger configured to receive a second stream of flue gas from the turbine and one of the stream of gaseous hydrogen or the stream of gaseous oxygen not received by the third heat exchanger, wherein the fourth heat exchanger is configured to transfer heat between the flue gas and one of the stream of gaseous hydrogen or the stream of gaseous oxygen to superheat one of the stream of hydrogen or the stream of oxygen; wherein the third heat exchanger and the fourth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; and wherein the system is configured to direct the stream of superheated oxygen and the stream of superheated hydrogen to the combustion chamber.
3. The system of claim 2, further comprising a fuel cell, and wherein the system is configured to: direct a first portion of the stream of gaseous hydrogen to the third heat exchanger or the combustion chamber; and direct a second portion of the stream of gaseous hydrogen to the fuel cell.
4. The system of claim 3, wherein the system is configured to direct the second portion of the stream of gaseous hydrogen to the fuel cell via the second heat exchanger.
5. The system of claim 3, wherein the system is configured to direct a stream of residual hydrogen from the fuel cell to the combustion chamber.
6. The system of claim 3, wherein the system is configured to direct a stream of residual hydrogen from the fuel cell to the third heat exchanger to superheat the stream of residual hydrogen, and wherein the system is configured to direct the stream of superheated residual hydrogen to the combustion chamber.
7. The system of claim 3, wherein the system comprises a fifth heat exchanger configured to receive a third stream of flue gas from the turbine and the stream of residual hydrogen from the fuel cell, wherein the fifth heat exchanger is configured to transfer heat between the third stream of flue gas and the stream of residual hydrogen to superheat the stream of residual hydrogen; wherein the third heat exchanger, the fourth heat exchanger, and the fifth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; and wherein the system is configured to direct the stream of superheated residual hydrogen to the combustion chamber.
8. The system of claim 7, further comprising: a water recovery heat exchanger configured to receive a fourth stream of flue gas from the turbine and a stream of ambient air, wherein the water recovery heat exchanger is configured to transfer heat from the fourth stream of flue gas to the

stream of ambient air to yield a stream of water; wherein the water recovery heat exchanger is arranged to receive the fourth stream of flue gas from at least one of the third heat exchanger, the fourth heat exchanger, or the fifth heat exchanger; and wherein the system is configured to direct the stream of water to a cooling system for the combustion chamber and/or the turbine.

**9.** A method comprising: directing a first stream of compressed air and a stream of liquid hydrogen to a first heat exchanger; transferring heat, using the first heat exchanger, between the first stream of compressed air and the stream of liquid hydrogen to yield a stream of liquid oxygen and a stream of gaseous hydrogen; directing the stream of liquid oxygen to a second heat exchanger; transferring heat, using the second heat exchanger, to the stream of liquid oxygen to yield a stream of gaseous oxygen; directing a first stream of flue gas from a turbine of a gas turbine and at least one of the stream of gaseous hydrogen or the stream of gaseous oxygen to a third heat exchanger; transferring heat, using the third heat exchanger, between the first stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; and directing the stream of hydrogen and the stream of oxygen to a combustion chamber of the gas turbine, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

**10.** The method of claim 9, wherein: the first stream of flue gas from the turbine of the gas turbine and one of the stream of gaseous hydrogen or the stream of gaseous oxygen are directed to the third heat exchanger; and the method further comprises: directing a second stream of flue gas from the turbine of the gas turbine and one of the stream of gaseous hydrogen or the stream of gaseous oxygen not directed to the third heat exchanger to a fourth heat exchanger, wherein the third heat exchanger and the fourth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; transferring heat, using the third heat exchanger, between the first stream of flue gas and the stream of gaseous hydrogen or the stream of gaseous oxygen to superheat the stream of hydrogen or the stream of oxygen; transferring heat, using the fourth heat exchanger, between the second stream of flue gas and the stream of gaseous oxygen or the stream of gaseous hydrogen to superheat the stream of oxygen or the stream of hydrogen; and directing the stream of superheated oxygen and the stream of superheated hydrogen to the combustion chamber of the gas turbine.

**11.** The method of claim 10, further comprising: directing a first portion of the stream of gaseous hydrogen from the first heat exchanger to the third heat exchanger or the combustion chamber; and directing a second portion of the stream of gaseous hydrogen from the first heat exchanger to a fuel cell.

**12.** The method of claim 11, wherein the second portion of the stream of gaseous hydrogen is directed from the first heat exchanger to the fuel cell via the second heat exchanger.

**13.** The method of claim 12, further comprising: directing a stream of residual hydrogen from the fuel cell to the combustion chamber.

**14.** The method of claim 12, further comprising: directing a stream of residual hydrogen from the fuel cell to the third heat exchanger; transferring heat, using the third heat exchanger, between the first stream of flue gas and the stream of residual hydrogen to superheat the stream of residual hydrogen; and directing the stream of superheated residual hydrogen to the combustion chamber.

**15.** The method of claim 12, further comprising: directing a third stream of flue gas from the turbine and the stream of residual hydrogen from the fuel cell to a fifth heat exchanger, wherein the third heat exchanger, the fourth heat exchanger, and the fifth heat exchanger are arranged to receive the streams of flue gas from the turbine in series or in parallel; transferring heat, using the fifth heat exchanger, between the third stream of flue gas and the stream of residual hydrogen to superheat the stream of residual hydrogen; and directing the stream of superheated residual hydrogen to the combustion chamber.

**16.** The method of claim 15, further comprising: directing a fourth stream of flue gas from the turbine and a stream of ambient air to a water recovery heat exchanger, wherein the water recovery



heat exchanger is arranged to receive the stream of flue gas from at least one of the third heat exchanger, the fourth heat exchanger, and the fifth heat exchanger; transferring heat, using the water recovery heat exchanger, from the fourth stream of flue gas to the stream of ambient air to yield a stream of water; and directing the stream of water to a cooling system for the combustion chamber and/or the turbine.

**17.** A system comprising: a gas turbine having a combustion chamber and a turbine, the gas turbine configured to receive a stream of gaseous oxygen and a stream of gaseous hydrogen and to output a stream of flue gas; and a water recovery heat exchanger configured to receive the stream of flue gas from the turbine, wherein the water recovery heat exchanger is configured to cool the stream of flue gas to yield a stream of water; wherein the system is configured to direct the stream of water to a cooling system for the combustion chamber and/or the turbine.

**18.** The system of claim 17, further comprising: a superheating heat exchanger configured to receive the stream of flue gas from the turbine and at least one of the stream of gaseous hydrogen or the stream of gaseous oxygen, wherein the superheating heat exchanger is configured to transfer heat between the stream of flue gas and the stream of gaseous hydrogen and/or the stream of gaseous oxygen to superheat the stream of hydrogen and/or the stream of oxygen; wherein the system is configured to direct the stream of flue gas to the water recovery heat exchanger via the superheating heat exchanger; wherein the water recovery heat exchanger is configured to receive the stream of flue gas and a stream of ambient air and to transfer heat from the stream of flue gas to the stream of ambient air to yield the stream of water; and wherein the system is configured to direct the stream of hydrogen and the stream of oxygen to the combustion chamber, wherein at least one of the stream of hydrogen and the stream of oxygen is superheated.

**19.** A method comprising: combusting a stream of gaseous oxygen and a stream of gaseous hydrogen in a combustion chamber of a gas turbine; directing a stream of flue gas from a turbine of the gas turbine and a stream of ambient air to a water recovery heat exchanger, wherein the water recovery heat exchanger transfers heat from the stream of flue gas to the stream of ambient air to yield a stream of water; and directing the stream of water to a cooling system for the combustion chamber of the gas turbine and/or the turbine of the gas turbine.

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